

PROBLEM STATEMENT #26 | Smart Cities, Transport & Logistics

Peer-to-Peer Carpooling & Expense Splitter

1. Problem Context & Overview

A commuter ridesharing portal that matches drivers and passengers with overlapping daily routes, enforces detour tolerance thresholds, and calculates automated fuel cost share splits.

Target Stakeholders / Actors: *Rider Commuter, Driver Host*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.

Sample Acceptance Criteria: Pass: Route match confirmed and fuel cost split calculated; Fail: Detour exceeding tolerance accepted.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.